European Commission

Linking forest structure and function

Mirco Migliavacca

EEBIOMASS workshop
27/10/2022

Joint Research Centre

Outline

- Why are we interested in structure and fluxes interplay?
- Ecosystem structural variables (some) and relation with functions;
- Scale emergent properies (structure and functions interplay at the eddy covariance sites);
- Concluding remarks

Why are we interested in structure and fluxes interplay?

Eddy covariance flux networks

Global Change Biology

INVITED RESEARCH REVIEW | Free Access

How eddy covariance flux measurements have contributed to our understanding of *Global Change Biology*

Dennis D. Baldocchi

First published: 28 August 2019 | https://doi.org/10.1111/gcb.14807 | Citations: 110

CSIRO PUBLISHING
www.publish.csiro.au/journals/ajb *Australian Journal of Botany*, 2008, **56**, 1–26

TURNER REVIEW No. 15

'Breathing' of the terrestrial biosphere: lessons learned from a global network of carbon dioxide flux measurement systems

Dennis Baldocchi

Ecosystem Sciences Division, Department of Environmental Science, Policy and Management, University of California, 137 Mulford Hall, Berkeley, CA 94720, USA.
Email: Baldocchi@nature.berkeley.edu

ICOS Integrated Carbon Observation System

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Home › Data products ›

Drought-2018 ecosystem eddy covariance flux product for 52 stations in FLUXNET-Archive format

DOI: https://doi.org/10.18160/YVR0-4898

Citation: Drought 2018 Team and ICOS Ecosystem Thematic Centre: Drought-2018 ecosystem eddy covariance flux product for 52 stations in FLUXNET-Archive format, , doi:10.18160/YVR0-4898, 2020.

View in the data portal

- Carbon, water and energy cycling research
- Effects of climate and meteorological variability on ecosystem scale fluxes
- Phenology and ecosystem functions
- Earth System Modelling and data driven upscaling
- Validation satellite products
- ….and many others

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How do fluxes relates with vegetation properties?

- “Measurements of meteorological conditions, physiological status of the vegetation, structure of the vegetation and state of the soil are crucial to interpret temporal and spatial variation in fluxes, and to relate fluxes with ecological traits” (Baldocchi et al., 2014, Global Change Biology)
- The interplay between vegetation properties, canopy/ecosystem structure and fluxes is still a partly unexplored research topic

Global Change Biology

Opinion | Free Access

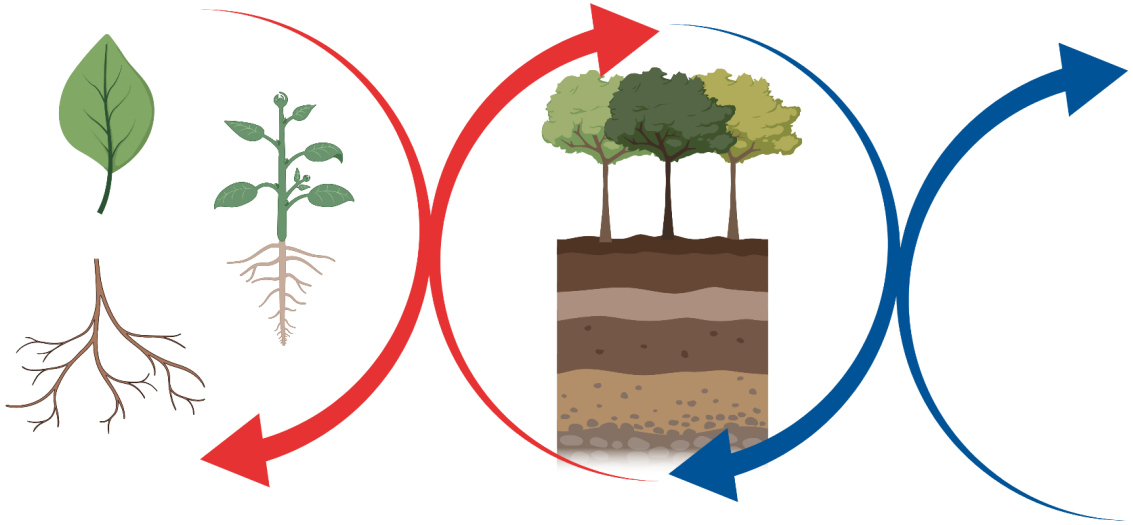
Measuring fluxes of trace gases and energy between ecosystems and the atmosphere – the state and future of the eddy covariance method

Dennis Baldocchi

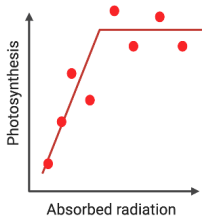
First published: 03 June 2014 | https://doi.org/10.1111/gcb.12649 | Citations: 274

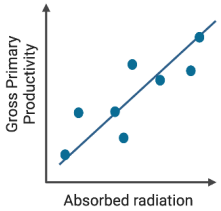
How do fluxes relates with vegetation properties?

- Complexity causes properties to emerge at the landscape scale, which do not operate at the cell, leaf, and plant scales;
- Eddy flux measurements are adept at discovering **scale emergent properties**
- How the functioning of the whole system differs from the sum of the individual parts (Baldocchi et al., 2014, Global Change Biology)



Organs level processes

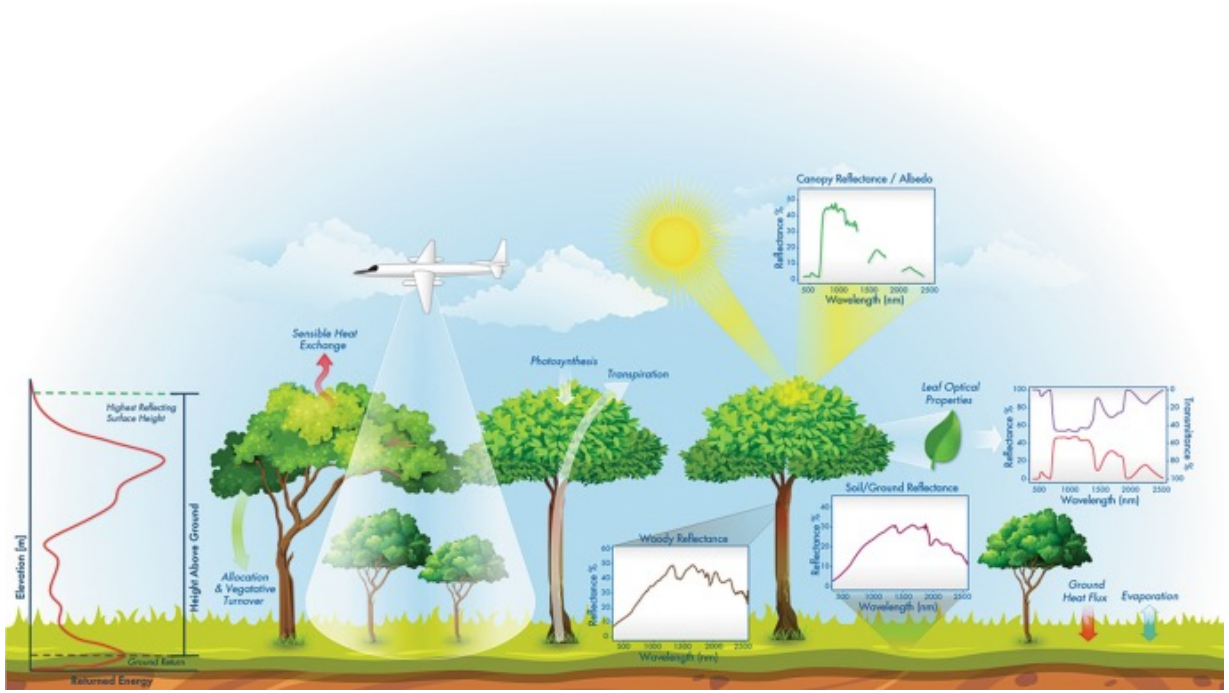




Emerging ecosystem fluxes

Canopy structure

Canopy structure key to grasp scale emergent properties

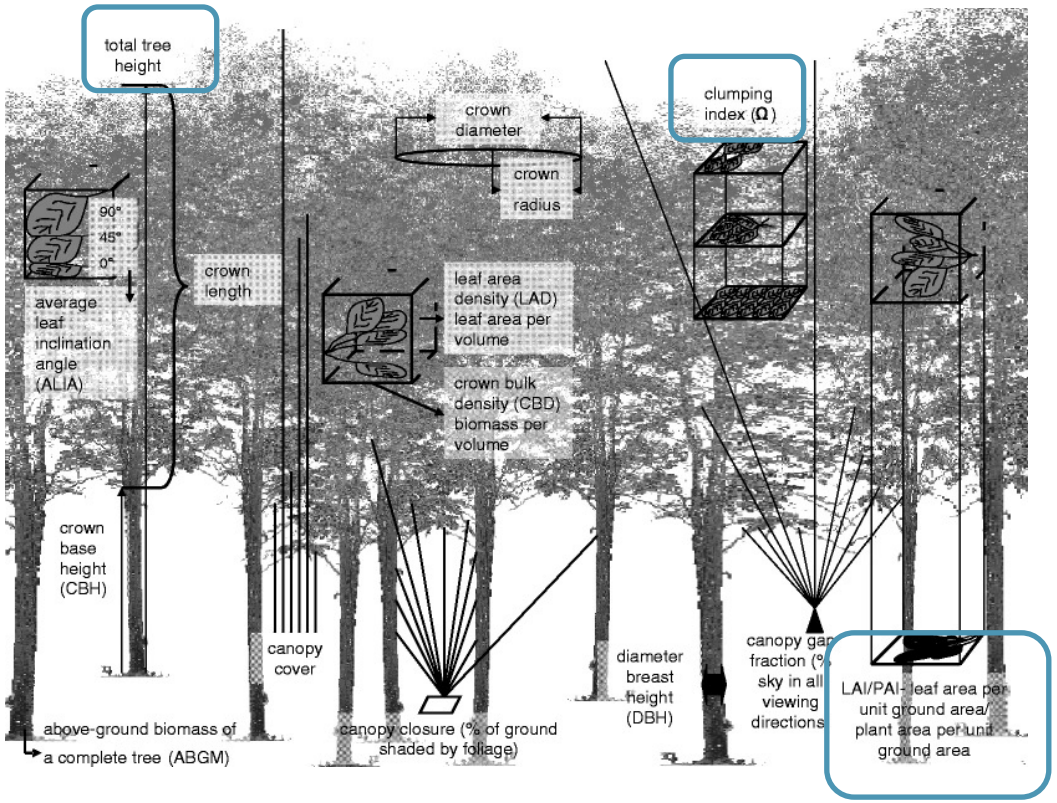


Scaling Functional Traits from Leaves to Canopies

Shawn P. Serbin & Philip A. Townsend

Chapter | Open Access | First Online: 23 June 2020

12k Accesses | 13 Citations



Review Paper | Open Access | Published: 18 March 2011

Review of ground-based methods to measure the distribution of biomass in forest canopies

Dominik Seidel, Stefan Fleck, Christoph Leuschner & Tom Hammett

Annals of Forest Science **68**, 225–244 (2011) | Cite this article

5713 Accesses | **52** Citations | **1** Altmetric | Metrics

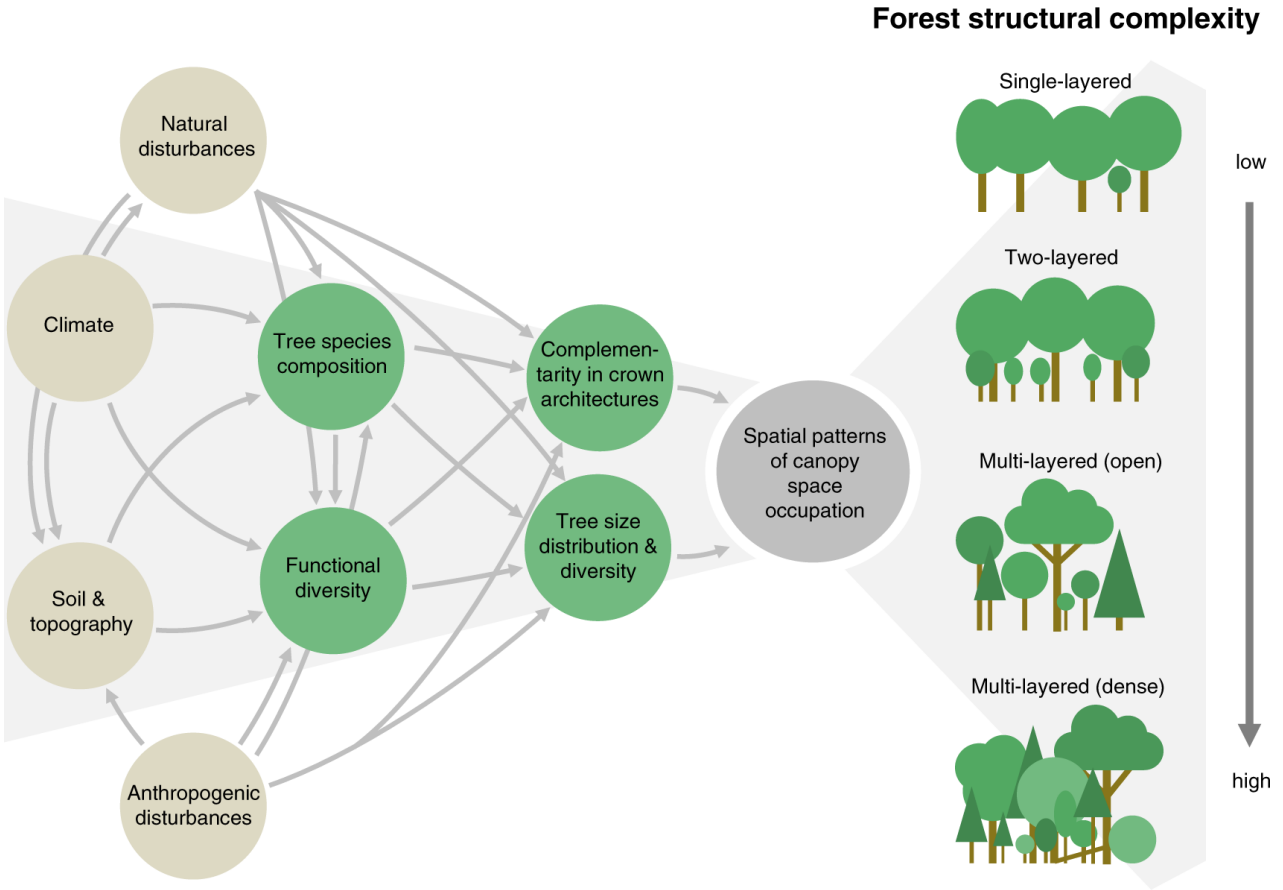
Drivers of forest (and canopy) structure

- Climate and soil characteristics
- Disturbances
- Management (humans)
- Monitoring structure in a consistent way is critical to understand ecosystem functions and surface fluxes

Article | Open Access | Published: 22 January 2021

Global patterns and climatic controls of forest structural complexity

Martin Ehbrecht, Dominik Seidel, Peter Annighöfer, Holger Kreft, Michael Köhler, Delphine Clara Zemp, Klaus Puettmann, Reuben Nilus, Fred Babweteera, Katharina Willim, Melissa Stiers, Daniel Soto, Hans Juergen Boehmer, Nicholas Fisichelli, Michael Burnett, Glenn Juday, Scott L. Stephens & Christian Ammer



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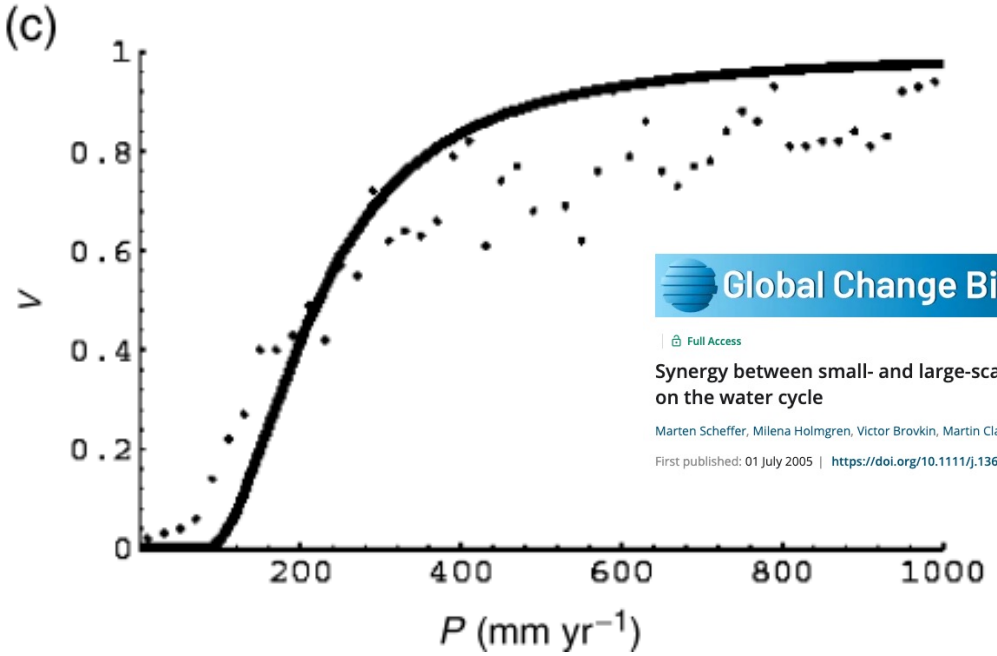
Few examples

- Leaf Area Index, leaf size, shape and orientation
  - Influence radiative transfer within the canopy (reflectance, interception and transmittance of light)
  - Sunlit/shaded leaves
  - Leaf boundary layer (with influence of decoupling, see Jarvis et al., 1996)
  - Leaf and needle clumping;
  - LAI is critically determined by soil water available (e.g. Scheffer et al., 2005), but controls through light absorption the main processes determining Carbon and water exchange, energy balance and

Source: https://land.copernicus.eu/global/products/lai



Global Change Biology

Full Access

Synergy between small- and large-scale feedbacks of vegetation on the water cycle

Marten Scheffer, Milena Holmgren, Victor Brovkin, Martin Claussen

First published: 01 July 2005 | https://doi.org/10.1111/j.1365-2486.2005.00962.x | Citations: 98

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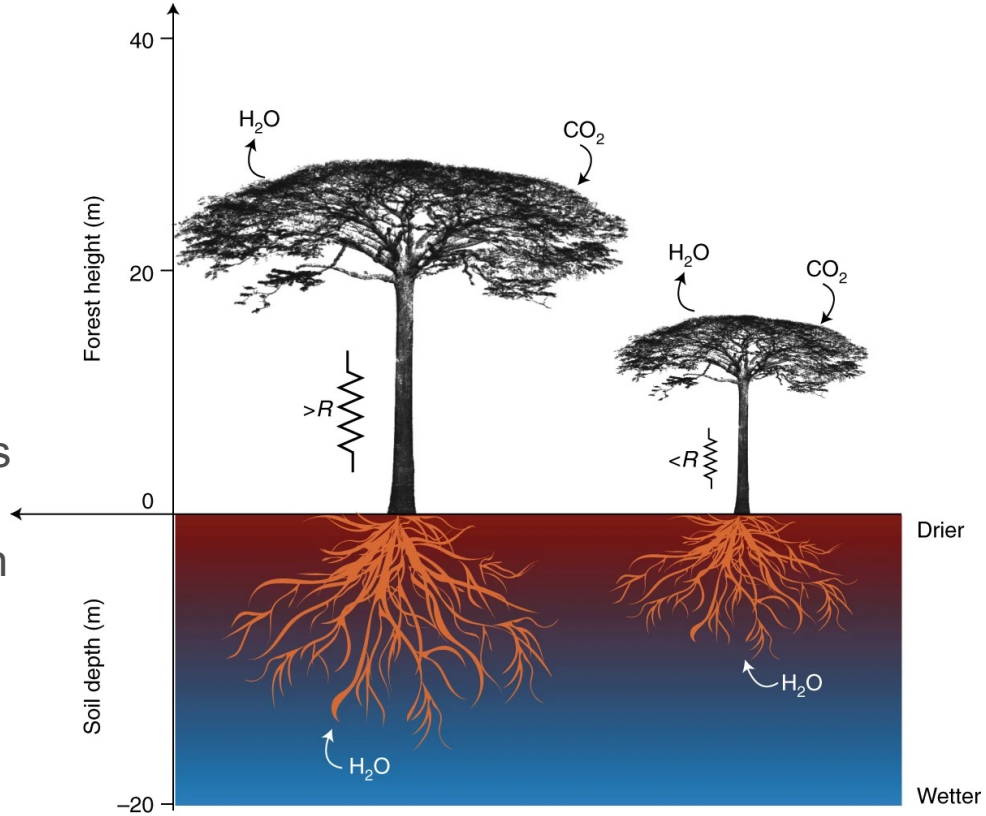
Few examples

- Tree / vegetation height
  - Determined by resource availability (hydraulic limitation hypothesis) and light competition
  - It controls the coupling between stomata and atmosphere by influencing surface roughness and then aerodynamic resistance
  - Reflects variation in water-use efficiency that decreases as a consequence of progressive hydraulic constraints on stomatal conductance to water vapour and growth in taller vegetation;
  - Important for seed dispersion patterns
  - Partly related with tree mortallity



Plant, Cell & Environment

Free Access

The hydraulic limitation hypothesis revisited

MICHAEL G. RYAN, NATHAN PHILLIPS, BARBARA J. BOND

First published: 14 March 2006 | https://doi.org/10.1111/j.1365-3040.2005.01478.x | Citations: 443

FOREST ECOLOGY

Tree height matters

Paulo Brando

Nature Geoscience **11**, 390–391 (2018) | Cite this article

1421 Accesses | **11** Citations | **29** Altmetric | Metrics

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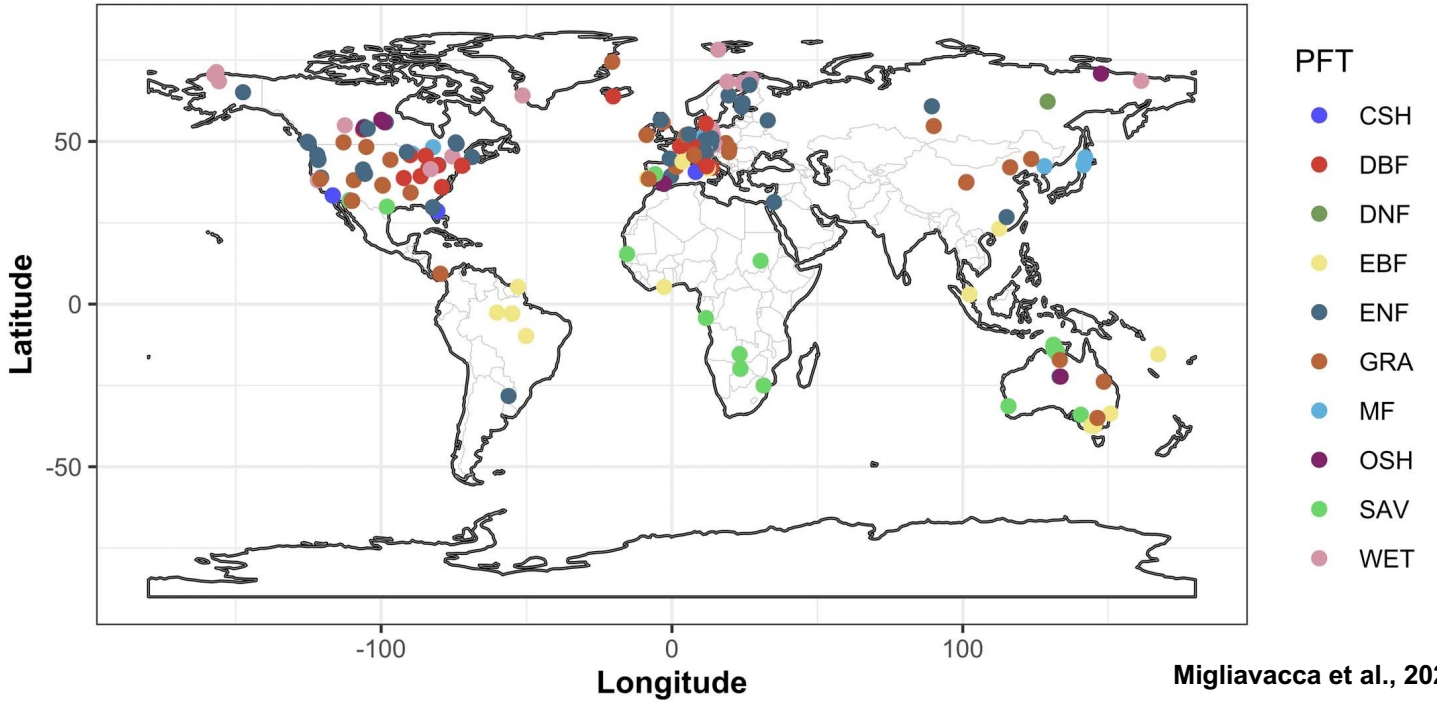
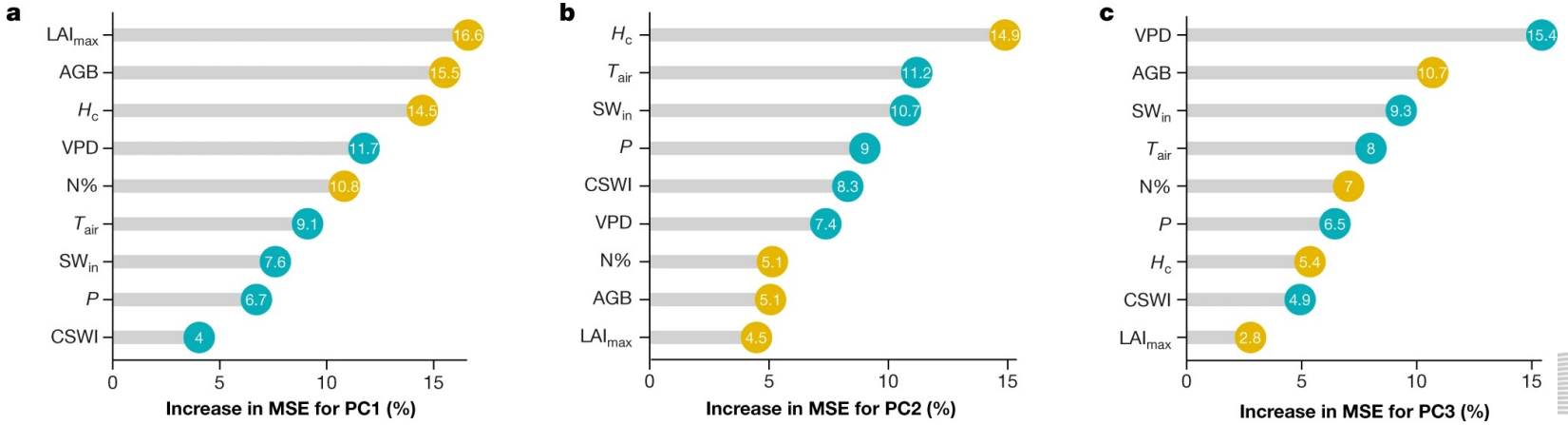
How does structure relates to functions in space



Migliavacca et al., 2021 Nature



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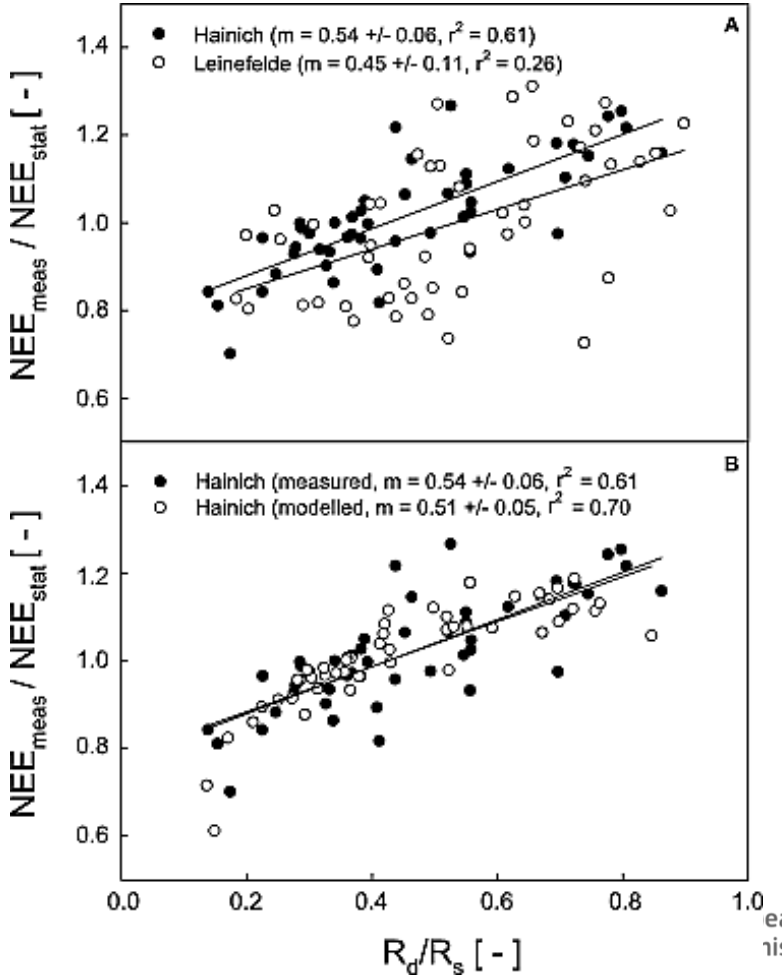
Examples of scale emergent properties

Photosynthesis – radiation relationship

- Photosynthesis does not always saturate at the canopy scale
  - Diffuse radiation affects the canopy gas exchange processes in a forest ecosystem
  - Increase in diffuse radiation or in general the leaf in shade operate in the linear part of the light response curve
  - LAI, clumping index, sunlit shaded leaved determine radiative regimes, and therefore relationship between net or gross CO2 uptake and light is modifies by strauctu re

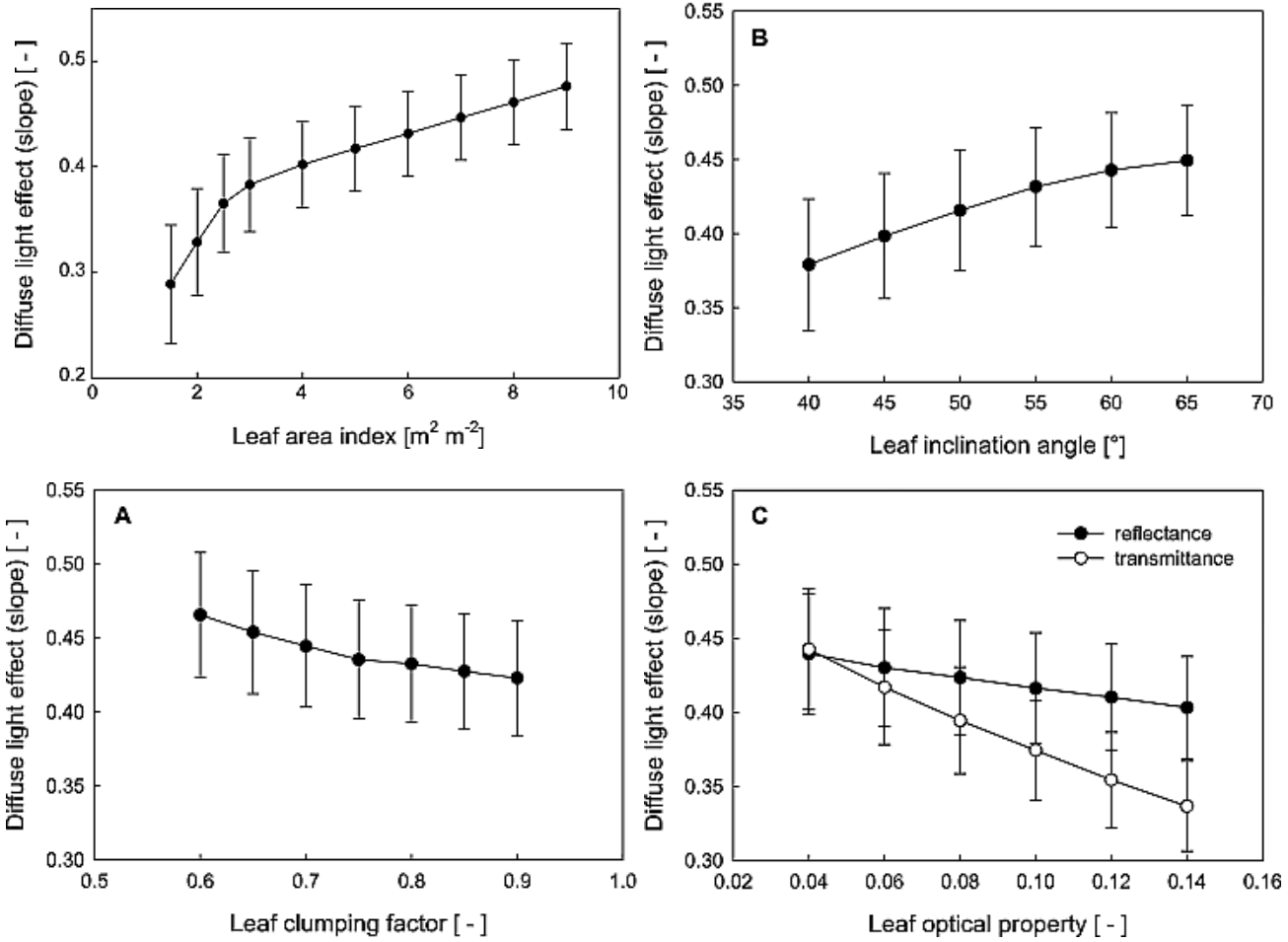
Knohl et al., Journal of Geophysical Research: Biogeosciences, Volume: 113, Issue: G2, First published: 13 May 2008, DOI: (10.1029/2007JG000663)



Photosynthesis – radiation relationship

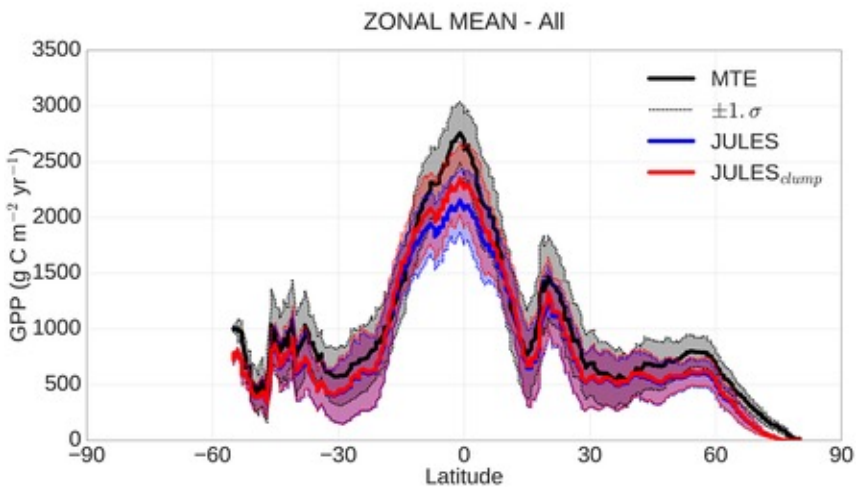
Knohl et al., Journal of Geophysical Research: Biogeosciences, Volume: 113, Issue: G2, First published: 13 May 2008, DOI: (10.1029/2007JG000663)

- This scale emergent properties is determined by structure
- LAI, clumping index, optical properties of leaved determine the radiative regimes
- Relationship between net or gross CO_2 uptake and light is modified by structure



…implications for Earth System Models

- Better description of structure and structural diversity is one key aspect to include in dynamic vegetation models to improve predictions of carbon fluxes:



Global Biogeochemical Cycles®

Research Article | Free Access

Underestimation of Global Photosynthesis in Earth System Models Due to Representation of Vegetation Structure

R. K. Braghiere, T. Quaife, E. Black, L. He, J. M. Chen

First published: 11 October 2019 | https://doi.org/10.1029/2018GB006135 | Citations: 17

…implications for modelling C, water fluxes

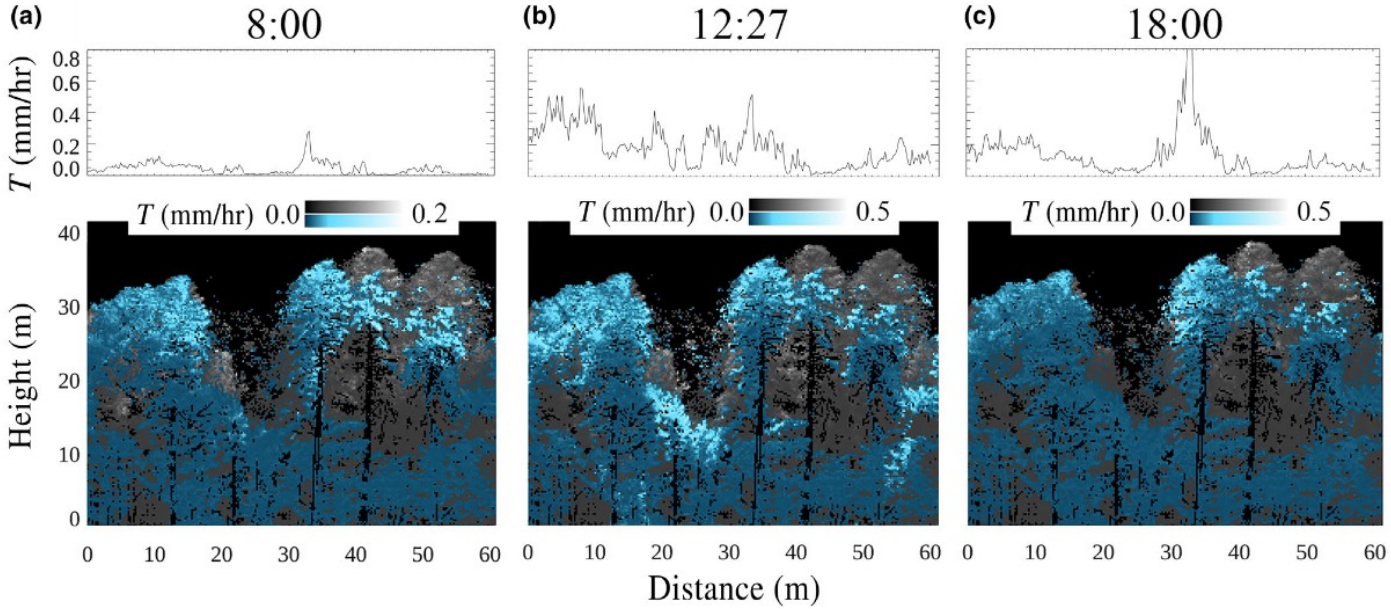
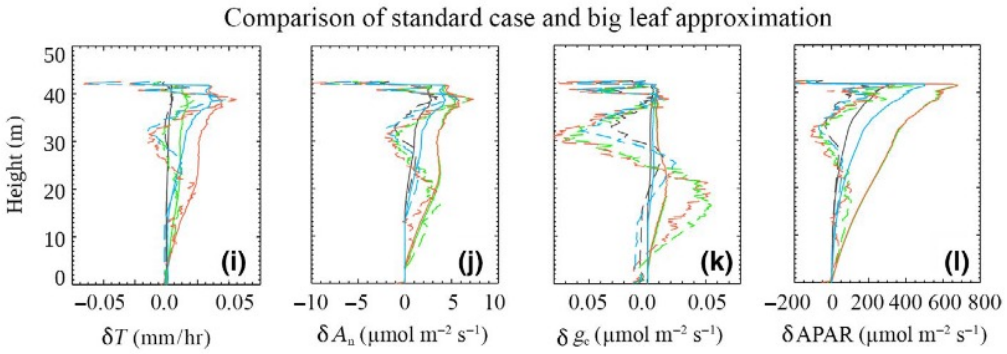


FIGURE 4 3D representation of forest transpiration (*T*). Panels show *T* at 8:00 l.t. (a), 12:27 l.t. (b) and 18:00 l.t. (c) under a water-saturated regime. The top graphs illustrate the horizontal *T*-profile of the blue coloured canopy slice shown below. The bottom graph illustrates the distribution of *T* for a specific slice (blue colours), while dark blue indicates low and bright blue indicates high *T* values. The grey coloured canopy represents the background trees with dark colours for low *T* and bright colours for high *T*. Sky and stems are coloured black. Please note that the bottom graphs have different scales, as indicated in the respective legends [Colour figure can be viewed at wileyonlinelibrary.com]



Remote sensing of forest gas exchange: Considerations derived from a tomographic perspective

Alexander Damm[1,2] | Eugenie Paul-Limoges[1] | Daniel Kükenbrink[1] | Christoph Bachofen[3] | Felix Morsdorf[1]

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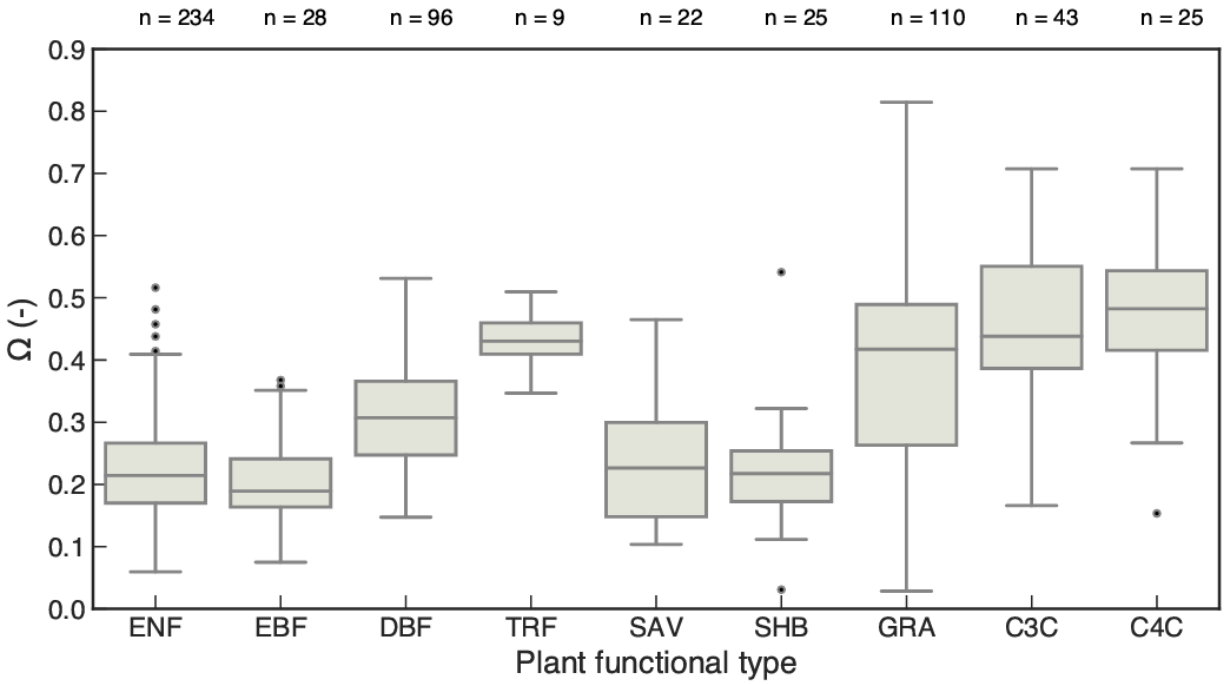
Transpiration sensitivity to stomatal conductace

- Transpiration sensitivity to stomatal conductance
- Function of the degree of coupling between the vegetation and the atmosphere: decoupling factor (Ω)
- See Jervis and McNaughton, 1986
- Possible impacts on WUE estimates and T/ET
- Depends critically on leaf size, canopy height, leaf length



Ideas and perspectives: how coupled is the vegetation to the boundary layer?

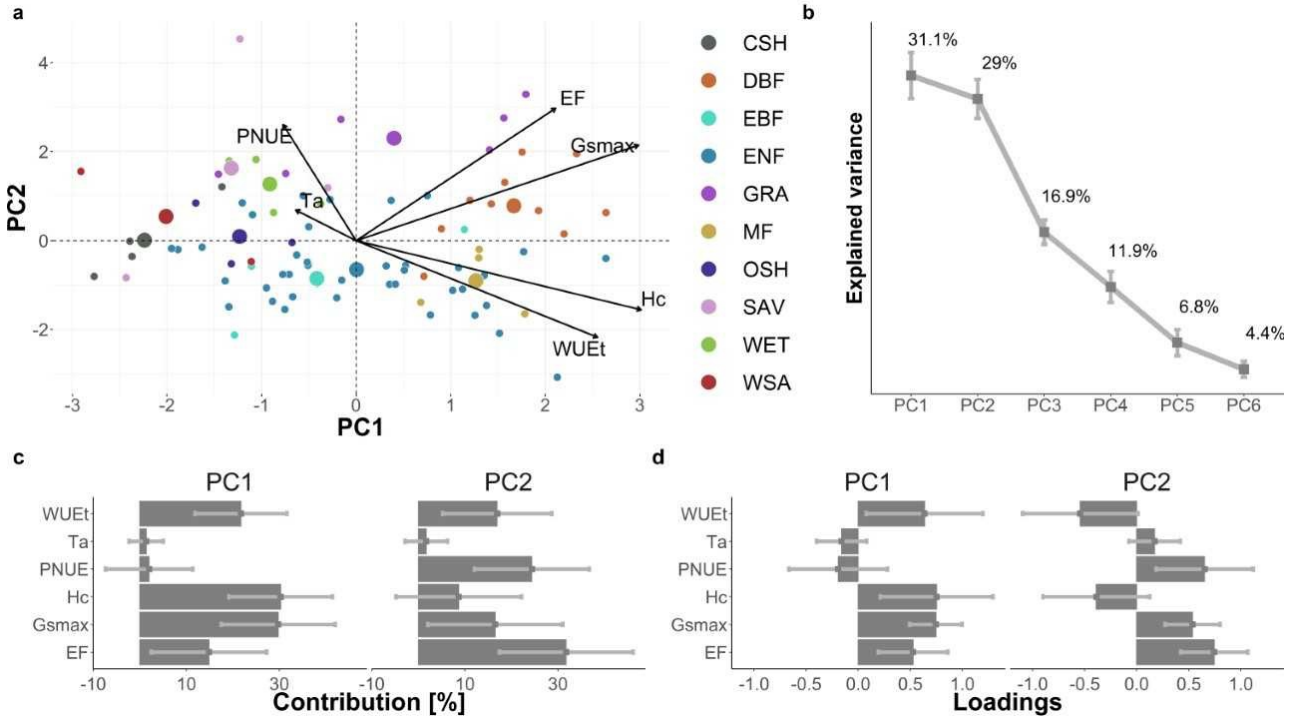
Martin G. De Kauwe[1,2], Belinda E. Medlyn[3], Jürgen Knauer[4], and Christopher A. Williams[5]

[1]ARC Centre of Excellence for Climate Extremes, University of New South Wales, Sydney, NSW 2052, Australia
[2]Department of Biological Science, Macquarie University, North Ryde, NSW 2109, Australia
[3]Hawkesbury Institute for the Environment, Western Sydney University, Locked Bag 1797, Penrith, NSW 2751, Australia
[4]Department of Biogeochemical Integration, Max Planck Institute for Biogeochemistry, 07745 Jena, Germany
[5]Graduate School of Geography, Clark University, 950 Main Street, Worcester, MA 01602, USA

Leaf traits coordination theories at the ecosystem scale

- Least cost hypothesis
- Water use efficiency (WUE) and photosynthetic Nitrogen Use efficiency (PNUE) are negatively related at the leaf level
- At the ecosystem scale this trade off is emerging only as secondary axis

Gomarasca et al., in preparation



Concluding remarks

Concluding remarks

- “Eddy flux measurements are adept at discovering **scale emergent properties”** (Baldocchi et al., 2014), that are critical to better understand and model surface atmosphere interactions

- Monitoring consistently across flux network ecosystem/vegetation structure (e.g. EO products) is a key step to better understand **scale emergent properties**

Thank you

Mirco Migliavacca

mirco.migliavacca@ec.europa.eu
Twitter: @BAIEbgi

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